RouteRanker Presentation

RouteRanker — Stakeholder Presentation

(12 slides. Speaker notes in italics.)

Slide 1 — Title

RouteRanker: AI-Powered Public Transit Route Optimization

Bengaluru BMTC case study — 4,416 routes analyzed

Slide 2 — The Problem

Indian city bus networks suffer from:

- 🚌 Overcrowded trunk routes at peak hours
- 🚌 Underused feeder routes off-peak
- 🚌 Duplicate coverage on shared corridors
- 🚌 No scalable method for planners to compare route efficiency

Today, route review is manual and quarterly at best. RouteRanker automates it.

Slide 3 — Approach

Input: GTFS timetable data (routes, trips, stop times, stops)

Output: Route-level efficiency labels + actionable recommendations

Method: Feature engineering → clustering → classification → rule-based optimizer → dashboard

We don’t need passenger counters to find inefficiencies — the timetable itself contains the signal.

Slide 4 — Data

Item | Value |

|---|---|

City | Bengaluru (BMTC) |
Routes | 4,416 |
Trips | 57,836 |
Stop events | 1,540,686 |
Stops | 9,960 |

The same pipeline runs unchanged on Delhi, Mumbai, Hyderabad, Pune — any GTFS feed.

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Slide 5 — What We Learned

- Trip counts are heavily right-skewed: median 3 trips/route, max 398

- 68% of routes are **balanced**

- 27% are **underutilized** (near-empty off-peak)

- 4% are **overcrowded** (peak-hour trunk corridors)

The 4% overcrowded routes serve the busiest corridors — they are where interventions matter most.

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Slide 6 — Method: Clustering

- **KMeans (k=3)** on 31 standardized features

- **Silhouette = 0.796** → excellent cluster separation

- **DBSCAN** flags 429 outlier routes for manual review

Three naturally distinct route archetypes emerge from the data — no manual grouping needed.

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Slide 7 — Method: Classification

- **XGBoost** and **RandomForest** predict route performance class

- Test accuracy: **99.8–100%**

- Feature importances align with planner intuition: trip_count, peak_ratio, route_length_km dominate

The model distills a hand-crafted business rule into a fast surrogate — planners can apply it to any new GTFS feed in seconds.

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Slide 8 — Optimization Engine

For each route, RouteRanker recommends:

Class | Recommendation |

|—|—|

Overcrowded | +15% peak frequency, larger buses, consider splitting |
Underutilized | −15% off-peak frequency, evaluate merging, mini-buses |
Balanced | Maintain, monitor quarterly |

Slide 9 — Simulation Results

Applying the optimizer to the current schedule:

- **184** overcrowded routes gain peak capacity

- **1,198** underutilized routes free up off-peak trips

- Net effect: **~3,000 trips/week redistributed** toward underserved corridors

- **No increase in total fleet size**

Slide 10 — The Dashboard

Streamlit app with 5 tabs:

1. **Overview** — KPIs and distribution charts

2. **Map** — interactive stop network across Bengaluru

3. **Predictions** — route-by-route performance table

4. **Optimization** — per-route recommendations

5. **Simulation** — current vs optimized trip counts

Planners can filter by class and cluster, drill into any route, and download CSVs for further analysis.

Slide 11 — Architecture & Extensibility

GTFS → clean → features → cluster + classify → optimize → dashboard

- **Plug-and-play** for any GTFS city feed

- **Modular** — swap models or add features without rewriting the pipeline

- **Fast** — full run in under 5 minutes on a laptop

Slide 12 — Roadmap

Next 3 months:

- Integrate APC (Automatic Passenger Counter) data for real ridership labels

- Add `shapes.txt`-based route overlap detection for merging recommendations

- Deploy containerized version with scheduled weekly GTFS refresh

Next 6 months:

- Multi-objective optimization (frequency + fleet + equity)

- Live road-network travel times via Google/Mapbox APIs

- Multi-city deployment: Delhi, Mumbai, Hyderabad

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Slide 13 — Ask / Next Steps

- **Pilot** with BMTC planning team on 20 routes

- **Data partnership** for APC/ticketing feeds

- **Budget** for cloud deployment + monitoring

RouteRanker is production-ready today on any GTFS feed. The only missing piece is real ridership data to convert proxies into ground truth.